Text 1

Specialisation can be seen as a response to the problem of an increasing accumulation of scientific knowledge. By splitting up the subject matter into smaller units, one man could continue to handle the information and use it as the basis for further research. But specialisation was only one of a series of related developments in science affecting the process of communication. Another was the growing professionalisation of scientific activity.

No clear-cut distinction can be drawn between professionals and amateurs in science: exceptions can be found to any rule. Nevertheless, the word “amateur” does carry a connotation that the person concerned is not fully integrated into the scientific community and, in particular, may not fully share its values. The growth of specialisation in the nineteenth century, with its consequent requirement of a longer, more complex training, implied greater problems for amateur participation in science. The trend was naturally most obvious in those areas of science based especially on a mathematical or laboratory training, and can be illustrated in terms of the development of geology in the United Kingdom.

A comparison of British geological publications over the last century and a half reveals not simply an increasing emphasis on the primacy of research, but also a changing definition of what constitutes an acceptable research paper. Thus, in the nineteenth century, local geological studies represented worthwhile research in their own right; but, in the twentieth century, local studies have increasingly become acceptable to professionals only if they incorporate, and reflect on, the wider geological picture. Amateurs, on the other hand, have continued to pursue local studies in the old way. The overall result has been to make entrance to professional geological journals harder for amateurs, a result that has been reinforced by the widespread introduction of refereeing, first by national journals in the nineteenth century and then by several local geological journals in the twentieth century.

As a logical consequence of this development, separate journals have now appeared aimed mainly towards either professional or amateur readership. A rather similar process of differentiation has led to professional geologists coming together nationally within one or two specific societies, whereas the amateurs have tended either to remain in local societies or to come together nationally in a different way.

Although the process of professionalisation and specialisation was already well under way in British geology during the nineteenth century, its full consequences were thus delayed until the twentieth century. In science generally, however, the nineteenth century must be reckoned as the crucial period for this change in the structure of science.

Text 2

A great deal of attention is being paid today to the so-called digital divide — the division of the world into the info (information) rich and the info poor. And that divide does exist today. My wife and I lectured about this looming danger twenty years ago. What was less visible then, however, were the new, positive forces that work against the digital divide. There are reasons to be optimistic.

There are technological reasons to hope the digital divide will narrow. As the Internet becomes more and more commercialized, it is in the interest of business to universalize access — after all, the more people online, the more potential customers there are. More and more governments, afraid their countries will be left behind, want to spread Internet access. Within the next decade or two, one to two billion people on the planet will be netted together. As a result, I now believe the digital divide will narrow rather than widen in the years ahead. And that is very good news because the Internet may well be the most powerful tool for combating world poverty that we've ever had.

Of course, the use of the Internet isn't the only way to defeat poverty. And the Internet is not the only tool we have. But it has enormous potential.

To take advantage of this tool, some impoverished countries will have to get over their outdated anti-colonial prejudices with respect to foreign investment. Countries that still think foreign investment is an invasion of their sovereignty might well study the history of infrastructure (the basic structural foundations of a society) in the United States. When the United States built its industrial infrastructure, it didn't have the capital to do so. And that is why America's Second Wave infrastructure-including roads, harbors, highways, ports and so on-were built with foreign investment. The English, the Germans, the Dutch and the French were investing in Britain's former colony. They financed them. Immigrant Americans built them. Guess who owns them now? The Americans. I believe the same thing would be true in places like Brazil or anywhere else for that matter. The more foreign capital you have helping you build your Third Wave infrastructure, which today is an electronic infrastructure, the better off you're going to be. That doesn't mean lying down and becoming fooled, or letting foreign corporations run uncontrolled. But it does mean recognizing how important they can be in building the energy and telecom infrastructures needed to take full advantage of the Internet.

Text 3

Why do so many Americans distrust what they read in their newspapers? The American Society of Newspaper Editors is trying to answer this painful question. The organization is deep into a long self-analysis known as the journalism credibility project.

Sad to say, this project has turned out to be mostly low-level findings about factual errors and spelling and grammar mistakes, combined with lots of head-scratching puzzlement about what in the world those readers really want.

But the sources of distrust go way deeper. Most journalists learn to see the world through a set of standard templates (patterns) into which they plug each day's events. In other words, there is a conventional story line in the newsroom culture that provides a backbone and a ready-made narrative structure for otherwise confusing news.

There exists a social and cultural disconnect between journalists and their readers which helps explain why the "standard templates" of the newsroom seem alien to many readers. In a recent survey, questionnaires were sent to reporters in five middle size cities around the country, plus one large metropolitan area. Then residents in these communities were phoned at random and asked the same questions.

Replies show that compared with other Americans, journalists are more likely to live in upscale neighborhoods, have maids, own Mercedeses, and trade stocks, and they're less likely to go to church, do volunteer work, or put down roots in community.

Reporters tend to be part of a broadly defined social and cultural elite, so their work tends to reflect the conventional values of this elite. The astonishing distrust of the news media isn't rooted in inaccuracy or poor reportorial skills but in the daily clash of world views between reporters and their readers.

This is an explosive situation for any industry, particularly a declining one. Here is a troubled business that keeps hiring employees whose attitudes vastly annoy the customers. Then it sponsors lots of symposiums and a credibility project dedicated to wondering why customers are annoyed and fleeing in large numbers. But it never seems to get around to noticing the cultural and class biases that so many former buyers are complaining about. If it did, it would open up its diversity program, now focused narrowly on race and gender, and look for reporters who differ broadly by outlook, values, education, and class.

Text 4

The world is going through the biggest wave of mergers and acquisitions ever witnessed. The process sweeps from hyperactive America to Europe and reaches the emerging countries with unsurpassed might. Many in these countries are looking at this process and worrying: "Won't the wave of business concentration turn into an uncontrollable anti-competitive force?"

There's no question that the big are getting bigger and more powerful. Multinational corporations accounted for less than 20% of international trade in 1982. Today the figure is more than 25% and growing rapidly. International affiliates account for a fast-growing segment of production in economies that open up and welcome foreign investment. In Argentina, for instance, after the reforms of the early 1990s, multinationals went from 43% to almost 70% of the industrial production of the 200 largest firms. This phenomenon has created serious concerns over the role of smaller economic firms, of national businessmen and over the ultimate stability of the world economy.

I believe that the most important forces behind the massive M&A wave are the same that underlie the globalization process: falling transportation and communication costs, lower trade and investment barriers and enlarged markets that require enlarged operations capable of meeting customers' demands. All these are beneficial, not detrimental, to consumers. As productivity grows, the world's wealth increases.

Examples of benefits or costs of the current concentration wave are scanty. Yet it is hard to imagine that the merger of a few oil firms today could re-create the same threats to competition that were feared nearly a century ago in the U.S., when the Standard Oil trust was broken up. The mergers of telecom companies, such as WorldCom, hardly seem to bring higher prices for consumers or a reduction in the pace of technical progress. On the contrary, the price of communications is coming down fast. In cars, too, concentration is increasing-witness Daimler and Chrysler, Renault and Nissan-but it does not appear that consumers are being hurt.

Yet the fact remains that the merger movement must be watched. A few weeks ago, Alan Greenspan warned against the megamergers in the banking industry. Who is going to supervise, regulate and operate as lender of last resort with the gigantic banks that are being created? Won't multinationals shift production from one place to another when a nation gets too strict about infringements to fair competition? And should one country take upon itself the role of "defending competition" on issues that affect many other nations, as in the U.S. vs. Microsoft case?